DSPY-Driven Automated Prompt Optimization for Structured Medical Triage

PES1UG24AM085 | PES1UG24AM091 |
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1. Why – Problem and Motivation

Medical triage determines the urgency of medical attention based on patient symptoms. Incorrect classification can cause delayed treatment (under-triage) or unnecessary use of healthcare resources (over-triage).

Large Language Models (LLMs) can generate triage predictions from symptom descriptions, but their outputs may vary with prompt design, leading to inconsistent or incomplete responses. Manually refining prompts is also time-consuming.

DSPy enables automated optimization of language-model programs using task-specific metrics. This project investigates whether DSPy can improve the accuracy and reliability of structured triage predictions compared with manually designed prompts.

2. What – Proposed Solution

We propose a system that uses DSPy to optimize medical triage prompts while keeping the underlying language model fixed. It accepts patient symptom descriptions and generates structured urgency predictions.

The solution involves:

1. **Dataset Preparation:** Select a synthetic or de-identified triage dataset with reference labels.
2. **Baseline:** Implement a manually designed prompt and measure its performance.
3. **Optimization:** Use DSPy and an optimizer such as MIPROv2 to refine instructions and examples.
4. **Prediction:** Generate structured triage outputs and validate their format and completeness.
5. **Evaluation:** Compare both approaches using accuracy, macro-F1, under-triage and over-triage rates, and output validity.

Scope: An experimental research prototype, not a clinical diagnostic tool or replacement for healthcare professionals.